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author: "Dr. William James Stanley"

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Millennium Argument Development Record

Audit note

This file preserves the development lineage rather than only the latest draft. The current conversation shows a repeated cycle: exploratory idea capture, synthesis, author amendment, reintegration, then publication branching. Precise per-message clock times are not available in the visible conversation record, so this document records the verified conversation date (2026-09-11) and does not fabricate finer timestamps.

Lineage map

| Stage | Date | Provenance | State | Material change |

|---|---|---|---|

| 0 | 2026-09-11 | User | Raw source / idea capture | Initial polished statement plus voice-typed amendment explaining emotional and philosophical motivation, self-comparison, systems-testing logic, IP concern, education concern, and Hodge context. |

| 1 | 2026-09-11 | AI synthesis from Stage 0 | First amalgamation | Converted raw material into a coherent personal essay while preserving bluntness, ancestry argument, systems-test analogy, and "I like shapes" origin. |

| 2 | 2026-09-11 | User direction + AI restructuring | Structured philosophical position | Reordered into explicit argumentative sections; protected private method details; separated scientific posture from anti-AI framing. |

| 3 | 2026-09-11 | Iterative user direction + AI rewrite | LinkedIn development | Shortened, added hook, domain vocabulary, philosophical framing, professional signalling, signature, and cohesive-paragraph rule. |

| 4 | 2026-09-11 | User direction | Publication split | Established two descendants: narrow LinkedIn post and broader website essay concept. |

| 5 | 2026-09-11 | User direction | Method abstraction | Writing process itself identified as a provenance-preserving research-development concept; documented separately and left incomplete. |

Stage 0 - user source material

OpenAI's decision to turn the Millennium Prize Problems into a large-scale capability test has bothered me all day, and I think I finally understand why.

It is not really about AI.

AI is a tool. I use it myself, including to study mathematics that is far beyond my formal training. I have been using the Hodge conjecture as a stress test-not because I seriously expect to solve it, but because trying to understand something that difficult exposes exactly where my own reasoning stops. It forces me backwards through definitions, assumptions, history and mathematics I do not yet know. That is the point. I am there to learn from the problem, not to conquer it.

What bothers me about OpenAI's Millennium exercise is something much older than AI: respect for the people who came before you.

Science has an ancestry.

You do not simply arrive at Navier-Stokes, Hodge, Riemann or any other great unresolved problem as though someone left a difficult puzzle sitting on a table. You arrive there because generations of people spent their lives constructing the intellectual world in which the problem can even be stated.

Someone invented the language. Someone proved the lemmas. Someone found the failed route that prevented everyone else wasting another decade on it. Someone noticed a strange relationship nobody else could see. Someone spent thirty years developing an idea that became one paragraph in somebody else's paper. Thousands of people taught the next generation enough mathematics to get them another few centimetres forward.

That is what an open problem contains.

It contains people.

So when a company hears that mathematicians may be making progress, mobilises enormous computational resources against the remaining Millennium problems, and then presents the outcome partly as a demonstration of what its models can accomplish, I find the posture difficult to admire.

Not because the computation is illegitimate. Not because the mathematics should somehow remain exclusively human. And certainly not because new tools should be kept away from old questions.

Because intellectual power does not exempt you from intellectual humility.

The extraordinary thing about a centuries-old scientific problem is not simply that nobody has managed to put the right symbols at the end of the page yet. It is that human beings have spent generations constructing an increasingly precise understanding of something reality refuses to surrender easily.

That accumulated understanding deserves more than being treated as the terrain for a technological victory lap.

A sufficiently powerful system may search mathematical space at a scale no individual researcher could approach. Good. That may become one of the greatest scientific instruments we have ever built.

But if it is going to participate in science, then it inherits the obligations of science too: provenance, attribution, criticism, patience, uncertainty, intellectual generosity, and respect for the work on which every apparent breakthrough stands.

Nobody reaches the frontier alone.

That includes AI companies.

And perhaps this is why my own little experiment with the Hodge conjecture feels so different to me. I am not standing outside the window trying to smash my way through because getting to the other side would look impressive.

I am standing at the window trying to understand who built it, why it has that shape, what I can actually see through it, and how much I still do not know.

There is nothing particularly glamorous about that.

But I recognise it as science.

Raw dictated amendment/capture from the same user turn:

Okay, so I'm going to try out this new thing where I just like recant the person to have that first. I want to see how like father's voice typing in finger and like if it like stresses it. Anyway, so this is the beginning. I have been really, really, really uptight and angry. Well, not angry, just pissed off the old egg. Like just something is just like nagging at me and I'm just like, why am I just in a really fucking bad mood? And I think I finally figured out what it is, it's because I found out yesterday that an AI has basically in my opinion committed line of the like biggest sins you can commit as a scientist, which is to just like take a very, very big like conceptually confounding like principle or

problem. And then just like, shove like a fucking robot onto it and just use it as like look at what technology is. Essentially like why I been like so annoyed lately with too fast and like technology going too fast okay what is your problem?

Like, I think that using, because it's a very logical thing to do for somebody who does like system testing and also works in logic, which I can assume urban AI is doing, to an extent. But like, it has for that is, hey, we have something good, it's like solving logic. What is the hardest logic problem, Millennium problem? That's what my thinking was, but I have respect for it because I have a history with it that will context. I respect the Millennium problems because I reckon it's excellent mathematics. They probably just sort as a problem with like throwing something out to prove a system. That's bad thinking and bad philosophy on how to do stuff. That's bad conduct. It's why I have a problem with education at the moment because everyone is just trying to use the simplest way to get around absolutely everything and not putting in the work. So yeah, I'm very upset that OpenAI for doing that, but I can't believe I completely pissed off at them because I am doing the same way, like sort of thing. But I'm doing it in a way that respects the problem. I'm doing it in a way that will teach me how to actually understand it better so I can then prove it or disprove it and actually come out on top and get the thing because I want to do this. This problem has stuck out to me for 10 years or longer because it's just about like shapes and shapes. I love shapes. And that's why it pisses me off so much that they're like somehow come up with a plan to like test something fairly similar to what I'm doing and then it's just like throwing in the face of everyone like I've been having I like hide behind a wall to do this. I like put up the most like some skewer security systems so nobody would know what I'm doing and then they just come out because they're a flipping company and don't have to do anything the way that the rest of us have to do anything to conserve IP because they have money and lawyers.

I think this amendment like addition of common strings puts on like my perspective properly of what I'm like thinking about the whole entire millennium thing in OpenAI. Can you now please like edit that so it makes sense? Because that's actually my voice.

Stage 1 - first synthesis / amalgamation

OpenAI's decision to turn the Millennium Prize Problems into a large-scale capability test has been bothering me all day, and I think I finally worked out why.

It is not really about AI.

AI is a tool. I use it myself. I use it to study mathematics that is miles beyond my formal training, and I have actually been using the Hodge conjecture as a kind of stress test.

Not because I seriously think I am going to casually solve the Hodge conjecture.

I use it because it is hard enough that I cannot bullshit my way through it.

Every time I think I understand something, the problem forces me backwards. What does that definition actually mean? Where did that assumption come from? What theorem is being used here? Why is this object defined that way? What mathematics do I need to learn before I am even entitled to have an opinion about this step?

That is the point.

I am using the problem to expose where my reasoning stops.

And maybe that is why OpenAI's Millennium exercise got under my skin so badly.

Because there is a very obvious systems-testing logic behind it.

You build a system that appears to be getting very good at formal reasoning. You want to find its limits. So you ask: what are some of the hardest reasoning problems humans have?

Well, there are the Millennium Prize Problems.

From a benchmarking perspective, I completely understand the move.

It is probably exactly the sort of thing I would think of.

But there is another part of this that matters to me much more, and it has almost nothing to do with AI.

It is about respect for the people who came before you.

Science has an ancestry.

You do not simply arrive at Hodge, Navier-Stokes, Riemann or any of the other great unresolved problems as though somebody left a particularly difficult puzzle sitting on a table and everyone has been too stupid to solve it.

Those problems exist inside intellectual worlds that generations of human beings spent their lives constructing.

Someone invented the language.

Someone proved the lemma that everybody now uses in three words.

Someone followed a dead end far enough to prove that nobody else needed to waste another decade doing it.

Someone noticed a relationship nobody had seen before.

Someone spent thirty years developing an idea that eventually became a paragraph in somebody else's paper.

Thousands of people taught the next generation enough mathematics to move the frontier another few centimetres.

That is what an open problem contains.

It contains people.

And this is where I think my irritation actually comes from.

There is something about taking one of those enormous, historically accumulated scientific problems, throwing industrial-scale computation at it, and then presenting the result partly as, "Look what our technology can do," that strikes me as intellectually backwards.

Not because the computation is illegitimate.

Not because mathematics should somehow remain exclusively human.

And definitely not because we should keep powerful new tools away from old scientific questions.

Use them.

Absolutely use them.

If AI becomes capable of exploring mathematical spaces at scales no individual researcher could ever approach, that could become one of the greatest scientific instruments we have ever built.

But intellectual power does not exempt you from intellectual humility.

If you are going to participate in science, you inherit the obligations of science as well.

Provenance.

Attribution.

Criticism.

Patience.

Uncertainty.

Intellectual generosity.

And respect for the work underneath whatever result you eventually produce.

Nobody arrives at the frontier alone.

That includes AI companies.

And there is another reason this hit me personally.

I have had the Hodge conjecture floating around in my head for probably a decade, for an extremely sophisticated mathematical reason:

I like shapes.

That is basically how it started.

There was something about the problem that grabbed me years ago, and eventually I decided that if I was going to keep thinking about it, I should actually learn enough mathematics to understand what the hell it was saying.

So my own little experiment with Hodge is also a systems test.

I work with logic. I build systems. Of course I eventually thought: if I want to test whether my way of reasoning is any good, why not point it at something brutally difficult?

So I cannot pretend the basic instinct is completely different.

It is not.

The difference, at least the one that matters to me, is what I think the problem is for.

I am not standing outside the window trying to smash my way through because getting to the other side would look impressive.

I am standing at the window trying to understand who built it.

Why it has that shape.

What mathematics makes the glass possible.

What I can actually see through it.

What I am merely imagining that I can see.

And how much I still do not know.

If one day that process somehow contributed something useful to the actual mathematics, fantastic.

But even if it never does, the problem has already done something valuable.

It has taught me where I am ignorant.

It has forced me to learn.

That matters to me because I think we are developing a much broader cultural problem with technology and education: increasingly, everything is treated as an obstacle to route around.

What is the fastest path?

What is the shortcut?

What tool will produce the output?

How do I get to the answer without having to do all of that annoying understanding in the middle?

And sometimes efficiency is exactly what we need.

But sometimes the work in the middle is the entire point.

There are things you should not want to bypass because the process of understanding them is what changes you.

That is true in education.

It is true in science.

And I think it should remain true even when our tools become extraordinarily powerful.

Maybe especially then.

Because if we build systems capable of reaching places that previously required generations of human work, the correct response should not be less respect for that history.

It should be more.

The machine may be new.

The frontier is not.

Stage 2 - structured philosophical argument

OpenAI's decision to direct substantial AI capability toward the remaining Millennium Prize Problems has bothered me since I heard about it.

It took me a while to understand why. The problem is not that AI is being used to do mathematics. I use AI myself, including to study mathematics far beyond my formal training. Nor do I think difficult scientific questions should somehow be protected from new computational tools. If artificial intelligence becomes capable of exploring mathematical structures at a scale no individual researcher could approach, that may become one of the most important scientific instruments ever created.

My objection is different. It concerns the posture we take toward problems that already have a scientific history.

A scientific problem is not merely a problem

The Millennium Prize Problems are often described as extraordinarily difficult mathematical questions. That is true, but incomplete. They are not isolated puzzles waiting for someone sufficiently clever-or sufficiently computationally powerful-to arrive and finish them. They are the visible frontier of enormous intellectual histories.

Before a problem such as the Hodge conjecture, the Riemann hypothesis, or Navier-Stokes can even be stated precisely, generations of mathematical work have already taken place. Someone developed the language in which the problem could be expressed. Someone established the definitions. Someone proved the foundational results. Someone followed a promising path until it failed and documented why. Someone discovered an unexpected relationship that changed how everyone else understood the field. Someone spent decades developing an idea that later appeared as a few lines inside another person's proof. Thousands of teachers and researchers transmitted that accumulated understanding to the people who moved the subject further.

An unsolved scientific problem therefore contains much more than an unanswered question. It contains an ancestry. It contains failed routes, partial successes, arguments, corrections, intuitions, disagreements, schools of thought and entire careers. It contains people.

Why the capability-test framing bothers me

From a systems-engineering perspective, I understand the temptation completely. Suppose you have developed a system that appears increasingly capable of mathematical reasoning. You want to discover its limits. A natural question follows: What are some of the hardest formal reasoning problems available? The Millennium Prize Problems are obvious candidates.

As a stress test, the logic is perfectly understandable. In fact, it is sufficiently obvious that I have independently used a related principle in my own work: take something difficult enough that superficial competence cannot survive contact with it, and use it to expose the boundary between apparent understanding and actual understanding.

The problem begins when the scientific object becomes primarily a demonstration of the system. There is an important difference between saying, "Here is a powerful new scientific instrument. Let us see whether it can contribute responsibly to a problem humans have studied for generations," and saying, in effect, "Here is one of humanity's hardest problems. Let us point our machine at it and see what that proves about the machine." Those may involve some of the same computations. They are not the same intellectual posture.

The second risks converting a scientific inheritance into a technological proving ground. That is what troubles me.

Power does not cancel obligation

The issue is not whether an AI system is entitled to participate in mathematics. Of course it is, insofar as it can produce useful mathematics. The deeper question is what obligations follow from participation in science.

Scientific work carries norms that developed for good reasons: provenance, attribution, criticism, uncertainty, reproducibility, patience, intellectual generosity, and respect for prior work. Those obligations do not disappear

because the new participant happens to be extraordinarily powerful. If anything, greater power makes them more important.

A system capable of searching mathematical possibility spaces far beyond ordinary human reach may genuinely discover things that individual researchers could not. That would be extraordinary. But even such a discovery would not have emerged from nowhere. The system would still be operating inside mathematical languages, concepts, theorems, conjectures, examples, counterexamples and representations constructed by human beings over centuries.

No matter how impressive the final step is, there is still a staircase underneath it. Nobody arrives at the frontier alone. That includes AI companies.

Why I think about this differently in my own work

This question is not abstract for me. For years I have been fascinated by the Hodge conjecture, initially for a very simple reason: I am strongly drawn to problems involving structure, geometry and relationships between different ways of representing the same thing.

I am not a professional algebraic geometer, and I do not pretend otherwise. What I have been doing is using an exceptionally difficult mathematical problem as a disciplined test of understanding. The details of that work are something I keep private, but the governing principle is straightforward.

I deliberately work against a problem difficult enough to reveal where my reasoning fails. When I encounter something I cannot justify, I go backwards. What does this definition mean? Why is this assumption permitted? What mathematical machinery is hidden inside this statement? What previous result makes this step possible? What am I treating as obvious simply because I do not yet know enough to see the difficulty?

The objective is not to force a route to an answer. The objective is to make ignorance visible. That distinction matters enormously to me. If I cannot understand the structures beneath the problem, then producing an answer by some shortcut would defeat the purpose of the exercise.

I am not using the problem because solving it would make my system look impressive. I am using the problem because its resistance makes weak reasoning difficult to hide. It is a teacher precisely because it refuses to cooperate.

The difference between reaching an answer and earning understanding

This connects to a broader concern I have about technology and education. We are becoming extremely good at removing friction. Often that is valuable. Some friction is pointless bureaucracy, repetition or mechanical labour, and eliminating it frees people to do better things. But not all difficulty is waste. Sometimes difficulty is where understanding is produced.

There is a growing temptation to treat every hard intermediate process as an inefficiency: What is the fastest route? What tool can produce the result? What can automate this step? How do we reach the output without having to traverse all of the uncomfortable reasoning in between?

Those questions are useful in engineering. They become dangerous when applied indiscriminately to learning and science. In some domains, the difficult middle is not an obstacle separating you from the valuable thing. The difficult middle is the valuable thing.

Learning mathematics is not merely obtaining mathematical answers. Scientific understanding is not merely producing correct outputs. There is a difference between possessing a result and understanding why the result belongs there. Technology can accelerate inquiry without abolishing that distinction.

What scientific humility requires

None of this means AI should withdraw from difficult mathematics. I want the opposite. I want these systems to become serious scientific instruments. I want them to identify connections researchers have missed, test conjectures at previously impossible scales, expose contradictions, generate candidate structures, and perhaps contribute to proofs that expand human knowledge.

But if they are going to enter science, then the institutions building them should approach science as participants rather than conquerors. That means acknowledging that the frontier was built before they arrived. It means distinguishing genuine mathematical contribution from demonstrations of model capability. It means preserving provenance. It means giving proper intellectual credit. It means making uncertainty visible. It means respecting failed work as part of the knowledge structure rather than treating only the final successful output as meaningful. And it means resisting the temptation to turn every great human question into a stage on which technological power can perform.

The Millennium Prize Problems are not sacred objects. They can be computed on. They can be attacked. They can be questioned. They can be solved by humans, machines, collaborations between the two, or perhaps by methods we have not imagined yet. But they deserve to be approached with some awareness of what they are.

They are not merely benchmarks. They are records of where human understanding has reached its current limit. That accumulated understanding deserves respect.

The principle underneath all of this

My objection, then, is not to artificial intelligence doing mathematics. It is to confusing capability with scientific maturity.

A powerful system may be able to reach further than we can. That does not automatically mean it understands the intellectual landscape through which it travelled, or that the institution deploying it has behaved responsibly toward that landscape. Scientific power and scientific humility are separate achievements. We should want both.

When I approach a problem such as the Hodge conjecture, I do not imagine myself standing outside a locked door trying to find the fastest way to break it open. I am trying to understand the door: who built it, why it has that particular structure, what had to be discovered before anyone could build it at all, why previous attempts to open it failed, what I can legitimately infer from what I can see, and what remains completely beyond my understanding.

Perhaps one day AI will help us open doors like that. I hope it does. But when it happens, the important story will not simply be that a powerful machine finally turned the handle. The story will include everyone who built the corridor leading to it.

Stage 3 - current LinkedIn skeleton / preferred technical register

The problem with using the Millennium Prize Problems as an AI benchmark is not the AI. It is the epistemology.

OpenAI's decision to direct large-scale model capability toward the remaining Millennium problems has bothered me more than I expected. Not because machine-assisted mathematics is illegitimate; quite the opposite. If AI becomes capable of traversing proof spaces, generating conjectural structure, testing formal consistency, or identifying latent relationships at scales inaccessible to individual mathematicians, it could become an extraordinary scientific instrument.

What concerns me is the framing. A Millennium problem is not merely a difficult search target in some abstract solution space. It is the exposed edge of a vast historical knowledge structure: definitions, failed proof strategies, partial results, category shifts, changes in representation, new abstractions, counterexamples, and entire mathematical languages built over generations.

That matters because science has ancestry. The Hodge conjecture, the Riemann hypothesis, and Navier-Stokes do not exist as isolated puzzles waiting for enough compute. They are downstream of centuries of conceptual work. Someone constructed the formalism. Someone proved the enabling lemma. Someone discovered that an apparently promising route was structurally impossible. Someone spent decades developing a framework that later became background notation.

So the distinction I care about is not human versus machine. It is **instrument versus spectacle**.

There is a substantial philosophical difference between asking, "Can this system make a legitimate contribution to an existing mathematical research programme?" and asking, "Can this system solve one of humanity's hardest problems and thereby demonstrate its capability?" The underlying computation may overlap. The scientific posture

does not.

I say this while doing something superficially similar myself. I have been using the Hodge conjecture as a stress test for my own reasoning and systems work. Not because I expect to brute-force my way toward a proof, and not because I think difficulty itself confers importance, but because the conjecture is sufficiently deep that weak understanding becomes visible very quickly.

When my reasoning fails, I move backwards through the dependency structure. What definition am I using? Which theorem licenses this inference? What assumptions are hidden in the representation? Am I confusing an algebraic object with its geometric interpretation? Have I mistaken a convenient category boundary for a real one? What prerequisite mathematics am I missing entirely?

That is the value of the exercise. It is not answer extraction. It is epistemic stress testing.

■ This is also why I am increasingly suspicious of our obsession with frictionless cognition. In education and technology, we often treat intermediate difficulty as inefficiency. Sometimes it is. But sometimes the difficult middle is where conceptual change occurs. Remove the friction and you may preserve the output while destroying the understanding that made the output meaningful.

AI may eventually contribute to a proof of one of these problems. I hope it does. But if AI is going to participate in science, then it inherits scientific obligations too: provenance, attribution, reproducibility, falsifiability where appropriate, uncertainty, criticism, and intellectual humility.

The frontier is not simply where the answer is missing. It is where accumulated human understanding currently stops.

Nobody reaches that frontier alone. Not mathematicians, not scientists, not models, and not the companies that build them.

Dr. William James Stanley

Stage 4A - LinkedIn publication branch

The Stage 3 text is currently the canonical LinkedIn branch. Its purpose is narrow: public controversy, scientific ancestry, instrument-versus-spectacle distinction, and the author's use of Hodge as epistemic stress testing. The broader education argument is not required in the eventual LinkedIn publication unless reintroduced by the author.

Stage 4B - website essay concept branch

Scientific Power Without Scientific Humility - website essay concept draft

This is deliberately **not** a finished essay. It captures the current philosophical spine while leaving the broader argument open for further author development.

OpenAI's use of the Millennium Prize Problems as a capability test raises a question that is easy to misstate as "AI versus mathematicians." That is not the question I care about. The interesting issue is what happens when a technology powerful enough to participate in scientific discovery enters an intellectual landscape that was built by generations of people before it arrived.

A major open problem is not simply an empty answer box. It is a historical structure: definitions, formalisms, partial results, failed proof strategies, counterexamples, changes in representation, and entire research programmes. Calling such a problem a benchmark is therefore never conceptually neutral. A benchmark asks what the problem can reveal about the system under test; science also asks what obligations the system and its operators have to the knowledge structure they are entering.

That distinction becomes clearer when I compare it with my own use of the Hodge conjecture. I use it as a deliberately difficult object against which to test understanding. The useful outcome is not merely getting closer to an answer. It is discovering where definitions fail, assumptions enter unnoticed, prerequisite mathematics is absent,

or a representation is being confused with the thing it represents. In that sense, the conjecture is functioning less like a trophy and more like an epistemic instrument.

The deeper argument I want to develop is about the difference between reducing friction and removing the conditions under which understanding forms. AI may eventually transform mathematics, and I want it to. But participation in science should bring scientific obligations with it: provenance, attribution, criticism, uncertainty, reproducibility where applicable, and intellectual humility.

The unresolved question for the essay is therefore not whether machines should solve difficult problems. It is what scientific maturity should mean when the entity capable of reaching the frontier is no longer a human researcher acting at human scale.

Open author questions

- What exactly distinguishes a scientific instrument from a capability spectacle?
- Which obligations belong to the model, which to the institution, and which to the scientific community using the system?
- How should provenance work when a result may emerge from training data, retrieval, generated intermediate reasoning, formal verification, and human correction?
- When does “friction removal” improve science, and when does it erase the epistemic work that makes understanding possible?
- How should historical scientific contribution be represented when machine-generated work depends on intellectual structures assembled across centuries?

Human gate: no conclusion is final. This document is a concept scaffold for further author development.

Provenance and authorship notes

- The underlying ideas, concerns, personal motivation, Hodge relationship, scientific-ancestry thesis, criticism of technological spectacle, education/friction concern, and IP boundary originate from the user's source material in this conversation.
- AI contributions in this lineage are editorial and structural: synthesis, ordering, compression, terminology selection, rhetorical framing, and publication adaptation.
- No AI-generated wording should be confused with independent evidence for claims about OpenAI or the Millennium Prize Problems. This record documents writing development, not factual verification of the underlying public event.
- The author retains the Human Gate over factual checking, final thesis, disclosure, and publication.